

FoC Guidelines

For Undergraduate Research Projects

SLIIT Faculty of Computing

(Effective from June 2024)



*** All undergraduate students in the Faculty of Computing must comply with the guidelines provided in this document before selecting a research project topic.*

PART 1 – PROJECT ALIGNMENT

1.1 Research Groups in FoC

Note: Your project must align with one of the research groups. Discuss this with your supervisors.

New Research Clusters	Technologies
Autonomous Intelligent Machines and Systems (AIMS)	Robotics (R), Embedded Systems (ES), Smart Systems (SS), Industrial IoT, real-time operations, Drone technology, Internet of Things (IoT), Industrial IoT, Wireless Communication, Sensor Network, Vehicular Networks
Computing Infrastructure (CI)	Software-defined networks (SDN), Wireless communication, Communication standards, Network Optimization, Integer Linear Programming ILP, Cloud Technologies, Sensor networks, Vehicular networks, Load balancing, Virtualization (NFV), Localization, Real packet inspection, Quality of Service (QoS), Internet of Things (IoT)
Information Assurance & Security (IAS)	Cyber security, Information Security (IS), Computer Networks (CN), Computer Governance (CG), IS Auditing (ISA), Blockchain, Data Privacy, Network security monitoring, and Application Security.
Software Systems & Technologies (SST)	Software Engineering (SE), High-Performance Computing (HPC), Parallel Computing (PC), Prog Languages (PL), Algorithms Design and Dev (ADD), Mobile Computing (MC), Rich Web-based Apps, Architecture and designing (RAAD)
Mixed Reality (MR)	Computer Graphics, Computer Vision, Augmented Reality (AR), Virtual Reality (VR)
Centre of Excellence for AI (CoEAI)	Machine Learning (ML), Deep Learning (DL), Soft Computing (SC), Image Processing (IP), Natural Language Processing (NLP), Computational Linguistics (CL), Data Science (DS), Business Intelligence and Analytics (BIA), Knowledge Management (KM), Big Data (BD), Intelligent recommender systems, Decision support systems.
Technology Integration & Management (TIM)	Information Systems Strategies & Policies, Technology Frameworks, Socio-economic Ecosystems, Business Process Management (BPM), Social Networks (SNA), Human-Computer Interaction (HCI), Usability Analytics (UA), User Acceptance, Requirement Engineering (RE), Ethical and social issues of technology integration, Applied research.

1.2 Sustainable Development Goals (SDGs)

Note: Your project must align with one of SDGs.

SDG Index	SDG Name	SDG Index	SDG Name
SDG 1	No Poverty	SDG 10	Reduced inequalities
SDG 2	Zero hunger	SDG 11	Sustainable cities and communities
SDG 3	Good health and well-being	SDG 12	Responsible consumption and production
SDG 4	Quality education	SDG 13	Climate action
SDG 5	Gender equality	SDG 14	Life below water
SDG 6	Clean water and sanitation	SDG 15	Life on land
SDG 7	Affordable and clean energy	SDG 16	Peace, justice, and strong institutions
SDG 8	Decent work and economic growth	SDG 17	Partnerships for the goals
SDG 9	Industry, innovation, and infrastructure		

1.3 Industry Vertical (Application Domain)

Note: Your project must align with one or more of the verticals

Industry Sectors / Verticals	
Agriculture (AgriTech)	Hospitality & Tourism
Chemical industries	Technology
Commerce & Business (BI, Digital Marketing etc.)	Law and Justice
Construction & Infrastructure	Livestock and Fisheries
Education	Media & Entertainment
Energy	Military
Environment and Climate (Forests, Oceans, Air, Wild Animals, Soil etc.)	Management (HRM, CRM, SRM, ERP, Procurement, Inventory etc.)
Fashion & Textiles	Manufacturing (Other)
Financial & Banking (FinTech)	Telecommunication
Food (Processing, Services)	Transportation & Logistics
Government/ Public services (SmartCity)	Veterinary services
Healthcare (Bio-medical, pharmaceutical)	Water Supply (Drinking, Irrigation)

PART 2 – Specialization Specific Requirements

2.1 RP (IT4010) Common Learning Outcomes.

Note: It is mandatory to fulfill these LOs to complete the research project successfully

LO1:	Provide a creative and innovative solution to an open-ended, complex problem.
LO2:	Apply the key pillars of the domain of computing.
LO3:	Apply project management methodologies to solve an open-ended problem using appropriate techniques, tools, best practices, and standards in the ethical, security, social, legal, and professional contexts.
LO4:	Communicate effectively with a wide range of audiences, including both technical and non-technical stakeholders.
LO5:	Assess the commercial viability of a project.

2.2 Program specific Research Areas (not limited to):

Note: Each component should focus on one or more of the program-specific research areas.

Information Technology (IT)	Machine Learning (ML), Deep Learning (DL), Software Engineering (SE), Mobile Applications, Human-Computer Interactions (HCI), Image Processing, Natural Language Processing (NLP), Algorithms, Operating Systems (OS), Cloud Computing (CC), Internet of Things (IoT), Database Systems (DS), e-learning
Information Systems Engineering (ISE)	Requirement Engineering (RE), Business Process Management (BPM), Project Management (PM), Usability Analytics, Social Networks Analytics (SNA), Social Media Analytics, Business Intelligence (BI), CRM/SRM/ERP, IS Strategy/ Policy Development, Technology Adoption & Diffusion, User Acceptance, Decision modeling, IS resource management, Visualizations, Human-Computer Interaction (HCI)
Software Engineering (SE)	Software Engineering (SE), High-Performance Computing (HPC), Parallel Computing (PC), Prog Languages (PL), Algorithms Design and Dev (ADD), Mobile Computing (MC), Architecture and designing (RAAD), Artificial Intelligence and ML, Visualizations
Cyber Security (CS)	Confidentiality, Integrity and Availability, Data & O/S Security, Cryptography, Secure Software Development, Information Security, Information security Audit, Risk Management, Governance and Compliances, Hardware/embedded systems/ IoT security, Vulnerability Assessment, Penetration Testing, Cyber Forensic Principles, Privacy Engineering, DevOps Security, Threat Hunting / Threat Intelligence, Mobile Security, Risk Automation, Social Engineering and Security, Network Security, Authentication SOAR
Data Science (DS)	Machine Learning (ML), Optimization Methods (OM), Big Data Analytics, Visual Analytics (VA), Statistical Modeling, Operational ML, Deep Learning (DL), Natural Language Processing (NLP), Computer Vision
Computer Systems and Network Engineering (CSNE)	Network Protocols and Technologies, Network Security, Network Programming, Virtualization and Cloud Computing, System Administration, Troubleshooting and Debugging and Optimization, Wireless Communication, IoT (Internet of Things), AI/ML for Networking, Database Administration, Computer Hardware and Microprocessor Systems, Network Management and Administration, Telecommunications Engineering, Network Design and Optimization, Software Defined Networking (SDN), Systems Security, Infrastructure Security, Cloud Security, Network standardization and Governance, Network Protocol, system Protocol, Networking API, Socket Programming, Secure Socket, Infrastructure Design and Development.
Interactive Media (IM)	AR, VR, Haptics and Immersive Experience Development, Game Design & Development, Interaction Design & UI/UX, CGI for Generic and Interactive Filmmaking, Computational Creativity with New Media Applications, Creative Coding for Design and Generative Arts, Digital Media Processing and Analysis, Interactive and Generative Story Telling Technologies, Computer Vision and 3D Object Representation, Animation and Motion Capturing, Metaverse Applications Development, Digital Marketing and Digital Business

2.3 Mandatory Submission Requirements of Different Specializations

Note:

- are the essential specialization-specific outputs that will be evaluated in addition to the submissions specified under the RP module.).
- Status1, Status2 discontinued. Introduce specialization checklists. Implement from 2024/25 June.

Summary of Specialization RP Checklists								
RP Schedule	Common Submissions	IT	SE	ISE	DS	CSNE	CS	IM
TAF								
Charter								
Proposal	Ethics Form							
PP 1 (50%)								
Status Doc1		Git Repository,	Git Repository	SRS Report, Business Model Canvas, Commercialization Plan video	Comprehensive Literature Review Report, Data Analysis Report	SRS/ DD/ PPR Report	SRS/ DD/ PPR Report	Hardware Feasibility Report, Video of Product Demo, PowerPoint Slides Commercialization Plan
PP 2 (90%)								
Status Doc 2	Evidence of using PM tool		UI/UX Demo Video, Deployment Report,	UAT Report,	ML-Ops Report, Ethical Analysis Document,	Performance Testing Report	Compliance Report	UAT Report/ Video
Final Evaluation (Demo, Prez, Viva)								
Thesis Submission								
Logbook	Reflective Statement							

Details about the RP checklists of different specializations

Check List Items	Why?	What should you do?	When to Submit?	How to Submit?	Submission Details
Information Technology Specialization					
GIT Repository	IT students should get experience in how repositories are used in industry projects. Evaluate version controlling and contribution from everyone	1. Create a git repository, including a README file. 2. An overview of the project, architectural diagram, and dependencies should be included in the README. 3. Full history of the project, including commits, branches, and merges	PP1	Upload to OneDrive. ("CheckList1" folder).	Shareable Git Repository Link with a README.MD file (<i>Group submission</i>)
Evidence of using a PM tool (MS Planner)	Train students on using a project management tool	1. Export report from PM tool.	PP2	Upload to OneDrive. ("CheckList2" folder).	Report (<i>Group submission</i>)
Software Engineering Specialization					
Git Repository	SE students must be able to demonstrate their skills in using GIT in their RP project, like how it has been used by the industry	1. Create a git repository with the README.MD file. 2. Use a branching model like GitHub flow to create branches. 3. Maintain the README.MD file with the major updates and breakthroughs of the project. 4. Include Architectural diagrams in the README.MD file.	PP1	Upload to OneDrive. ("CheckList1" folder)	Shareable Git Repository Link with a README.MD file (<i>Group submission</i>)
UI/UX Demo Video	SE students should present their product demo to the SE department to get feedback on the level of acceptability of the product	Prepare a video recording that includes the following: initial sketches, conceptual models, user feedback for the mockups, initial changes, finalized UI mockups, user feedback or heuristic analysis after finalizing the UI mockups, and a demonstration of the app. The video should be informative, with narration being optional, and it should not exceed 10 minutes.	PP2	Upload to OneDrive. ("CheckList2" folder).	Video (<i>Group submission</i>)
Deployment Report	SE students should be able to complete the RP project following the SDLC. This report will capture the process	1. Maintain this report from the start of the project. 2. Include the software specifications. 3. Include the details of the tools, package, third-party libraries, 4. Include the details of the deployment process. 5. Include the platform/server/cloud configuration details. 6. Include the test cases and test scripts.	PP2	Upload to OneDrive. ("CheckList2" folder). The department will provide a template.	Report (<i>Group submission</i>)
Evidence of using a PM tool (MS Planner)	Train students on using a project management tool	1. Export report from PM tool.	PP2	Upload to OneDrive. ("CheckList2" folder).	Report (<i>Group submission</i>)
Information Systems Engineering Specialization					

Business Model Canvas, Commercialization plan	To articulate the business concept, value proposition, target market, and marketing strategies. To ensure the business's stakeholders understand the project's direction and goals.	1. Complete the Business Model Canvas by Alexander Osterwalder based on your product requirements (Reference: https://www.strategyzer.com/library/the-business-model-canvas) 2. Create a 120-second video to showcase the commercialization plan based on the information presented on the Business Model Canvas.	PP1	Upload to OneDrive. ("CheckList1" folder). The department will provide a template.	Report, Video (Group submission)
Product Requirements Specification (SRS)	To capture the functional and non-functional requirements of the client or stakeholders regarding the software product.	1. Complete the SRS document, including the functional and non-functional requirements. (Ref: https://www.computer.org/education/bodies-of-knowledge/software-engineering?source=softwarerequirements) Include evidence of Requirement Elicitation, (E.g., Sfia framework) or evidence of BPR into the APPENDIX	PPI	Upload to OneDrive. ("CheckList1" folder). The department will provide a template.	Report + Appendix (Group submission)
User Acceptance Testing (UAT) Report	To ensure that the software meets the specified requirements and fulfills the needs of the end-users. Allow stakeholders to verify that the software behaves as expected and provides the intended functionality.	1. Identify a suitable client to implement the product. 2. Identify end-users of the system. 3. Create Test Scenarios, test steps, test date requirements, and expected results. 4. Evaluate Functional Requirements, User Interface (UI) Requirements, Usability Requirements, Compatibility Requirements, Security Requirements, Data Integrity Requirements, Regulatory Compliance Requirements and Documentation Requirements. 5. Document actual results and test pass/fail status and comments.	After PP2	Upload to OneDrive. ("CheckList2" folder). The department will provide a template.	Report (Group submission)
Evidence of using a PM tool (MS Planner)	Train students on using a project management tool	1. Export report from PM tool.	PP2	Upload to OneDrive. ("CheckList2" folder).	Report (Group submission)
Data Science Specialization					
Comprehensive Literature review	To ensure the understanding of the research scope and appropriate materials /approaches.	1. Perform a comprehensive study on the research area and identify appropriate techniques, tools, and methodologies.	Before proposal	Upload to OneDrive. ("CheckList1" folder). The department will provide a template.	Report (Group submission)
Data Analysis Report	To ensure the proper understanding of the selected datasets.	1. Generate analysis on selected datasets and identify their suitability for the research area.	PP1	Upload to OneDrive. ("CheckList1" folder). The department	Report (Group submission)

				will provide a template.	
MLOps Report	To ensure the appropriate understanding of best practices, sets of concepts, and a development culture regarding the end-to-end conceptualization, implementation, monitoring, deployment, and scalability of machine learning products.	1. Generate a report based on the template given by the department	PP2	Upload to OneDrive. ("CheckList2" folder). The department will provide a template.	Report (Group submission)
Ethical Analysis Report	To encourage the students to think about ethical aspects and reflect upon the primary purpose of using data and potential negative impacts.	1. Generate an ethical analysis document considering the ethical aspects of the research project, including datasets, models, and outcomes.	PP2	Upload to OneDrive. ("CheckList2" folder). The department will provide a template.	Report (Group submission)
Evidence of using a PM tool (MS Planner)	Train students on using a project management tool	1. Export report from PM tool.	PP2	Upload to drive ("CheckList2" folder).	Report (Group submission)

Computer Systems & Network Engineering Specialization

DD/SRS/PPR (Design Document for hardware designs OR SRS for software-related, OR PPR document.)	Depending on the project category, SRS/DD/PPR is needed to understand the student's ability to demonstrate the implementation plan and potential use cases.	1. Submit an SRS/DD/PPR document indicating the following: Security, Architecture Diagrams: Visual representations of the system or network architecture, highlighting controls, zones, trust boundaries, and data flows. 2. The DD can include technical drawings, circuit diagrams, and other related information. 3. PPR can include example frameworks, protocol design	PP1	Upload to OneDrive. ("CheckList1" folder). The department will provide a template.	Report (Group submission)
Performance Testing Report	The development should undergo a performance testing phase focused on the research component.	Performance Testing can be done based on a defined threshold or baseline. E.g. Throughput Packet Loss Prepare a detailed performance testing report with the test data/ scenario description.	After PP2	Upload to OneDrive. ("CheckList2" folder). The department will provide a template.	Report (Group submission)
Evidence of using a Project Management tool (MS Planner)	Train students on using a project management tool	1. Export report from PM tool.	PP2	Upload to OneDrive. ("CheckList2" folder).	Report (Group submission)

Cyber Security Specialization

DD/ SRS/ PPR (Design Document for hardware designs OR SRS for software-related, OR	Depending on the project category, One of the SRS/DD/PPR is needed to understand the student's ability to demonstrate the implementation plan and potential use cases.	1. Submit SRS/DD/PPR document indicating the following, Security. 2. Architecture Diagrams: Visual representations of the system or network architecture, highlighting security controls,	PP1	Upload to OneDrive. ("CheckList1" folder). The department	Report (Group submission)
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PPR document.)		zones, trust boundaries, and data flows. These diagrams help in understanding the security posture of the system...///. 3. The document may also include content related to Threat Modelling.		will provide a template.	
Secure Development/ Coding Practices (Compliance Report)	Depending on the project category, students must demonstrate using secure coding practices and secure implementation procedures (in the case of hardware-based implementations).	1. Submit a document as Proof. The document shall include a few of the following depending on the project. 1. Coding Standards Adherence 2. Input Validation and Sanitization 3. Output Encoding 4. Authentication and Authorization 5. Secure Configuration 6. Error Handling 7. Secure Communication 8. Secure File Handling 9. Secure Cryptographic Practices 10. Code Review and Testing 11. Documentation of Vulnerabilities and Fixes	After PP2	Upload to OneDrive. ("CheckList2" folder). The department will provide a template.	Report (Group submission)
Evidence of using a Project Management tool (MS Planner)	Train students on using a project management tool	1. Export report from PM tool.	PP2	Upload to OneDrive. ("CheckList2" folder).	Report (Group submission)
Interactive Media Specialization					
Hardware Feasibility Report	To assess students' understanding of hardware constraints before project commencement.	Prepare a hardware feasibility report considering the required hardware such as VR headsets for VR apps and smart devices for AR apps. (Don't consider any devices that are developed in this project)	Proposal	Upload to OneDrive. ("CheckList1" folder). The department will provide a template.	Report (Group submission)
Commercialization plan for the product	Ensure the communication, product demonstration, and product presentation are clear to the product's end user.	1. Create a 2–5-minute video and presentation slides to demonstrate the product to a potential client or an investor.	PP1	Upload to OneDrive. ("CheckList1" folder). The department will provide a template.	Video (Group submission)
User Experience and Acceptance Testing (UAT) Report	To ensure that the software application meets the specified requirements or solves the problem and fulfills the needs of the end-users. Allow stakeholders to verify that the software behaves as expected and addresses the required functional and non-functional requirements.	1. Identify and develop suitable customer personas. 3. Create Test Scenarios, test steps, test data requirements, and expected results. 4. Evaluate Functional Requirements, User Interface (UI) Requirements, Usability Requirements, Compatibility Requirements, Security Requirements, Data Integrity Requirements, Regulatory Compliance Requirements and Documentation Requirements.	Final	Upload to OneDrive. ("CheckList2" folder). The department will provide a template.	Report (Group submission)

		5. Document results and test pass/fail status and comments.			
Evidence of using a Project Management tool (MS Planner)	Train students on using a project management tool	1. Export report from PM tool.	PP2	Upload to OneDrive. ("CheckList2" folder).	Report (Group submission)